2.2: An element's properties depend on the structure of its atoms

Each element consists of a certain type of atom that is different from the atoms of any other element. An **atom** is the smallest unit of matter that still retains the properties of an element. Atoms are so small that it would take about a million of them to stretch across the period printed at the end of this sentence. We symbolize atoms with the same abbreviation used for the element that is made up of those atoms. For example, the symbol C stands for both the element carbon and a single carbon atom.

Subatomic Particles

Although the atom is the smallest unit having the properties of an element, these tiny bits of matter are composed of even smaller parts, called *subatomic particles*. Using high-energy collisions, physicists have produced more than 100 types of particles from the atom, but only three kinds of particles are relevant here: **neutrons**, **protons**, and **electrons**. Protons and electrons are electrically charged. Each proton has one unit of positive charge, and each electron has one unit of negative charge. A neutron, as its name implies, is electrically neutral.

Protons and neutrons are packed together tightly in a dense core, or **atomic nucleus**, at the center of an atom; protons give the nucleus a positive charge. The rapidly moving electrons form a "cloud" of negative charge around the nucleus, and it is the attraction between opposite charges that keeps the electrons in the vicinity of the nucleus. **Figure 2.4** shows two commonly used models of the structure of the helium atom as an example.

Figure 2.4

Simplified models of a helium (He) atom.

Simplified models of a helium atom.

Figure a. A cloud of negative charge around a center with two brown spheres and two pink spheres with each pink sphere labeled with a plus sign. This model represents the two electrons as a cloud of negative charge, a result of their motion around the nucleus. A yellow haze surrounds the nucleus.

Figure b. A circle with two small yellow spheres each labeled with a minus sign. The center of the circle has two brown spheres and two pink spheres with each pink sphere labeled with a plus sign. In this more simplified model, the electrons are shown as two small yellow spheres on a circle around the nucleus.

The helium nucleus consists of 2 neutrons (brown) and 2 protons (pink). Two electrons (yellow) exist outside the nucleus. These models are not to scale; they greatly overestimate the size of the nucleus in relation to the electron cloud.

The neutron and proton are almost identical in mass, each about 1.7×10−24 gram (g). Grams and other conventional units are not very useful for describing the mass of objects that are so minuscule. Thus, for atoms and subatomic particles (and for molecules, too), we use a unit of measurement called the **dalton**, in honor of John Dalton, the British scientist who helped develop atomic theory around 1800. (The dalton is the same as the *atomic mass unit*, or *amu*, a unit you may have encountered elsewhere.) Neutrons and protons have masses close to 1 dalton. Because the mass of an electron is only about 1/2,000 that of a neutron or proton, we can ignore electrons when computing the total mass of an atom.

Atomic Number and Atomic Mass

Atoms of the various elements differ in their number of subatomic particles. All atoms of a particular element have the same number of protons in their nuclei. This number of protons, which is unique to that element, is called the **atomic number** and is written as a subscript to the left of the symbol for the element. The abbreviation He2, for example, tells us that an atom of the element helium has 2 protons in its nucleus. Unless otherwise indicated, an atom is neutral in electrical charge, which means that its protons must be balanced by an equal number of electrons. Therefore, the atomic number tells us the number of protons and also the number of electrons in an electrically neutral atom.

We can deduce the number of neutrons from a second quantity, the **mass number**, which is the total number of protons and neutrons in the nucleus of an atom. The mass number is written as a superscript to the left of an element's symbol. For example, we can use this shorthand to write an atom of helium as He24. Because the atomic number indicates how many protons there are, we can determine the number of neutrons by subtracting the atomic number from the mass number. In our example, the helium atom He24 has 2 neutrons. For sodium (Na):

Superscript 22, sub 11, N a.

Mass number equals number of protons plus neutrons equals 23 for sodium. Atomic number equals number of protons equals number of electrons in a neutral atom equals 11 for sodium. Number of neutrons equals mass number minus atomic number equals 23 minus 11 equals 12 for sodium.

The simplest atom is hydrogen H11, which has no neutrons; it consists of a single proton with a single electron.

Because the contribution of electrons to mass is negligible, almost all of an atom's mass is concentrated in its nucleus. Neutrons and protons each have a mass very close to 1 dalton, so the mass number is close to, but

slightly different from, the total mass of an atom, called its **atomic mass**. For example, the mass number of sodium (Na1123) is 23, but its atomic mass is 22.9898 daltons; the difference is explained below.

Animation

Atomic Number and Atomic Mass

image

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Isotopes

All atoms of a given element have the same number of protons, but some atoms have more neutrons than other atoms of the same element and therefore have greater mass. These different atomic forms of the same element are called **isotopes** of the element. In nature, an element may occur as a mixture of its isotopes. As an example, the element carbon, which has the atomic number 6, has three naturally occurring isotopes. The most common isotope is carbon-12, C612, which accounts for about 99% of the carbon in nature. The isotope C612 has 6 neutrons. Most of the remaining 1% of carbon consists of atoms of the isotope C613, with 7 neutrons. A third, even rarer isotope, C614, has 8 neutrons. Notice that all three isotopes of carbon have 6 protons; otherwise, they would not be carbon. Although the isotopes of an element have slightly different masses, they behave identically in chemical reactions. (For an element with more than one naturally occurring isotope, the atomic mass is an average of those isotopes, weighted by their abundance. Thus, carbon has an atomic mass of 12.01 daltons.)

Both C12 and C13 are stable isotopes, meaning that their nuclei do not have a tendency to lose subatomic particles, a process called decay. The isotope C14, however, is unstable, or radioactive. A **radioactive isotope** is one in which the nucleus decays spontaneously, giving off particles and energy. When the radioactive decay leads to a change in the number of protons, it transforms the atom to an atom of a different element. For example, when a carbon-14 (C14) atom decays, a neutron decays into a proton, transforming the atom into a nitrogen (N14) atom. Radioactive isotopes have many useful applications in biology.

Radioactive Tracers

Radioactive isotopes are often used as diagnostic tools in medicine. Cells can use radioactive atoms just as they would use nonradioactive isotopes of the same element. The radioactive isotopes are incorporated into biologically active molecules, which are then used as tracers to track atoms during metabolism, the chemical processes of an organism. For example, certain kidney disorders are diagnosed by injecting small doses of radioactively labeled substances into the blood and then analyzing the tracer molecules excreted in the urine. Radioactive tracers are also used in combination with sophisticated imaging instruments, such as PET scanners that can monitor growth and metabolism of cancers in the body **(Figure 2.5)**.

Figure 2.5

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A PET scan, a medical use for radioactive isotopes.
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PET scan with a bright yellow spot in the throat indicating cancerous throat tissue.

PET, an acronym for positron-emission tomography, detects locations of intense chemical activity in the body. The bright yellow spot marks an area with an elevated level of radioactively labeled glucose, which in turn indicates high metabolic activity, a hallmark of cancerous tissue.

Although radioactive isotopes are very useful in biological research and medicine, radiation from decaying isotopes also poses a hazard to life by damaging cellular molecules. The severity of this damage depends on the type and amount of radiation an organism absorbs. One of the most serious environmental threats is radioactive fallout from nuclear accidents. The doses of most isotopes used in medical diagnosis, however, are relatively safe.

Radiometric Dating

EVOLUTION Researchers measure radioactive decay in fossils to date these relics of past life. Fossils provide a large body of evidence for evolution, documenting differences between organisms from the past and those living at present and giving us insight into species that have disappeared over time. While the layering of fossil beds establishes that deeper fossils are older than more shallow ones, the actual age (in years) of the fossils in each layer cannot be determined by position alone. This is where radioactive isotopes come in.

A “parent” isotope decays into its “daughter” isotope at a fixed rate, expressed as the **half-life** of the isotope —the time it takes for 50% of the parent isotope to decay. Each radioactive isotope has a characteristic half-life that is not affected by temperature, pressure, or any other environmental variable. Using a process called **radiometric dating**, scientists measure the ratio of different isotopes and calculate how many half-lives (in years) have passed since an organism was fossilized or a rock was formed. Half-life values range from

very short for some isotopes, measured in seconds or days, to extremely long—uranium-238 has a half-life of 4.5 billion years! Each isotope can best "measure" a particular range of years: Uranium-238 was used to determine that moon rocks are approximately 4.5 billion years old, similar to the estimated age of Earth. In the **Scientific Skills Exercise**, you can work with data from an experiment that used carbon-14 to determine the age of an important fossil. (Figure 25.6 explains more about radiometric dating of fossils.)

Scientific Skills Exercise

Calibrating a Standard Radioactive Isotope Decay Curve and Interpreting Data

AP Science Practices 1, 4, 5

How Long Might Neanderthals Have Co-Existed with Modern Humans (*Homo sapiens*)? Neanderthals (*Homo neanderthalensis*) were living in Europe by 350,000 years ago and may have coexisted with early *Homo sapiens* in parts of Eurasia for hundreds or thousands of years before Neanderthals became extinct. Researchers sought to more accurately determine the extent of their overlap by pinning down the latest date Neanderthals still lived in the area. They used carbon-14 dating to determine the age of a Neanderthal fossil from the most recent (uppermost) archeological layer containing Neanderthal bones. In this exercise you will calibrate a standard carbon-14 decay curve and use it to determine the age of this Neanderthal fossil. The age will help you approximate the last time the two species may have coexisted at the site where this fossil was collected.

How the Experiment Was Done Carbon-14(C14) is a radioactive isotope of carbon that decays to N14 at a constant rate. C14 is present in the atmosphere in small amounts at a constant ratio with both C13 and C12, two other isotopes of carbon. When carbon is taken up from the atmosphere by a plant during photosynthesis, C12, C13, and C14 isotopes are incorporated into the plant in the same proportions in which they were present in the atmosphere. These proportions remain the same in the tissues of an animal that eats the plant. While an organism is alive, the C14 in its body constantly decays to N14 but is constantly replaced by new carbon from the environment. Once an organism dies, it stops taking in new C14 but the C14 in its tissues continues to decay, while the C12 in its tissues remains the same because it is not radioactive and does not decay. Thus, scientists can calculate how long the pool of original C14 has been decaying in a fossil by measuring the ratio of C14 to C12 and comparing it to the ratio of C14 to C12 present originally in the atmosphere. The fraction of C14 in a fossil compared to the original fraction of C14 can be

converted to years because we know that the half-life of C14 is 5,730 years—in other words, half of the C14 in a fossil decays every 5,730 years.

Data from the Experiment The researchers found that the Neanderthal fossil had approximately 0.0078 (or, in scientific notation, 7.8×10−3) as much C14 as the atmosphere. The following questions will guide you through translating this fraction into the age of the fossil.

Excavated Neanderthal fossils.

Graph of radioactive isotope decay for Neanderthal fossils.

Data from R. Pinhasi et al., Revised age of late Neanderthal occupation and the end of the Middle Paleolithic in the northern Caucasus, *Proceedings of the National Academy of Sciences USA* 147:8611–8616 (2011). doi 10.1073/pnas.1018938108

The fraction of isotope remaining in the fossil is on the y-axis ranging from 0 to 1.0, with increments every tenth. Time before present (half-lives) is on the x-axis from 0 to 10, with increments every one whole unit. The data points are approximately (x, y): (0, 1), (1, 0.5), (2, 0.25), (3, 1.2), (4, 0.5) (5, 0.3) (6, 0.2), (7, 0.1), (8, 0.05), (9, 0.03), (10, 0).

INTERPRET THE DATA

- The graph shows a standard curve of radioactive isotope decay. The line shows the fraction of the radioactive isotope over time (before the present) in units of half-lives. Recall that a half-life is the amount of time it takes for half of the radioactive isotope to decay. Labeling each data point with the corresponding fractions will help orient you to this graph. Draw an arrow to the data point for half-life=1 and write the fraction of C14 that will remain after one half-life. Calculate the fraction of C14 remaining at each half-life and write the fractions on the graph near arrows pointing to the data points. Convert each fraction to a decimal number and round off to a maximum of three significant digits (zeros at the beginning of the number do not count as significant digits). Also write each decimal number in scientific notation.
- Recall that C14 has a half-life of 5,730 years. To calibrate the x-axis for C14 decay, write the time before present in years below each half-life.
- The researchers found that the Neanderthal fossil had approximately 0.0078 as much C14 as found originally in the atmosphere.
- Using the numbers on your graph, determine how many half-lives have passed since the Neanderthal died.
- Using your C14 calibration on the x-axis, what is the approximate age of the Neanderthal fossil in years (round off to the nearest thousand)?

- Approximately when did Neanderthals become extinct according to this study?
- The researchers cite evidence that modern humans (*H. sapiens*) became established in the same region as the last Neanderthals approximately 39,000–42,000 years ago. What does this suggest about possible overlap of Neanderthals and modern humans?
- Carbon-14 dating works for fossils up to about 75,000 years old; fossils older than that contain too little C14 to be detected. Most dinosaurs went extinct 65.5 million years ago.
- Can C14 be used to date dinosaur bones? Explain.
- Radioactive uranium-235 has a half-life of 704 million years. If it was incorporated into dinosaur bones, could it be used to date the dinosaur fossils? Explain.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology.**

The Energy Levels of Electrons

The simplified models of the atom in Figure 2.4 greatly exaggerate the size of the nucleus relative to that of the whole atom. If an atom of helium were the size of a typical football stadium, the nucleus would be the size of a pencil eraser in the center of the field. Moreover, the electrons would be like two tiny gnats buzzing around the stadium. Atoms are mostly empty space. When two atoms approach each other during a chemical reaction, their nuclei do not come close enough to interact. Of the three subatomic particles we have discussed, only electrons are directly involved in chemical reactions.

An atom's electrons vary in the amount of energy they possess. **Energy** is defined as the capacity to cause change—for instance, by doing work. **Potential energy** is the energy that matter possesses because of its location or structure. For example, water in a reservoir on a hill has potential energy because of its altitude. When the gates of the reservoir's dam are opened and the water runs downhill, the energy can be used to do work, such as moving the blades of turbines to generate electricity. Because energy has been expended, the water has less energy at the bottom of the hill than it did in the reservoir. Matter has a natural tendency to move toward the lowest possible state of potential energy; in our example, the water runs downhill. To restore the potential energy of a reservoir, work must be done to elevate the water against gravity.

The electrons of an atom have potential energy due to their distance from the nucleus **(Figure 2.6)**. The negatively charged electrons are attracted to the positively charged nucleus.

Figure 2.6

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Energy levels of an atom’s electrons.
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Atomic nucleus model with first shell, second shell, and third shell. Three arrows pointing away from the nucleus indicate energy absorbed and three arrows pointing toward the nucleus indicate energy lost.

Electrons exist only at fixed levels of potential energy called electron shells.

Atomic nucleus model with the first shell (lowest energy level), second shell (next highest energy level), and third shell (highest energy level in this model) depicted. Three arrows pointing away from the atomic nucleus indicate stepwise electron movement when energy is absorbed as follows: from the first to second shell; second to third shell, and first to third shell. Three arrows pointing towards the atomic nucleus indicate electron movement when energy is lost as follows: from the third to second shell; second to first shell, and third to first shell.

Figure 2.6 Walkthrough

Energy levels of an atom’s electrons

Watch an explanation of this figure in the Figure 2.6 Walkthrough: Energy levels of an atom's electrons.

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It takes work to move a given electron farther away from the nucleus, so the more distant an electron is from the nucleus, the greater its potential energy. Unlike the continuous flow of water downhill, changes in the potential energy of electrons can occur only in steps of fixed amounts. An electron having a certain amount of energy is something like a ball on a staircase (see Figure 2.6a). The ball can have different amounts of potential energy, depending on which step it is on, but it cannot spend much time between the steps. Similarly, an electron’s potential energy is determined by its energy level. An electron can exist only at certain energy levels, not between them.

An electron’s energy level is correlated with its average distance from the nucleus. Electrons are found in different **electron shells**, each with a characteristic average distance and energy level. In diagrams, shells can be represented by concentric circles, as they are in Figure 2.6b. The first shell is closest to the nucleus,

and electrons in this shell have the lowest potential energy. Electrons in the second shell have more energy, and electrons in the third shell even more energy. An electron can move from one shell to another, but only by absorbing or losing an amount of energy equal to the difference in potential energy between its position in the old shell and that in the new shell. When an electron absorbs energy, it moves to a shell farther out from the nucleus. For example, light energy can excite an electron to a higher energy level. (Indeed, this is the first step taken when plants harness the energy of sunlight for photosynthesis, the process that produces food from carbon dioxide and water.) When an electron loses energy, it "falls back" to a shell closer to the nucleus, and the lost energy is usually released to the environment as visible light or ultraviolet radiation.

Electron Distribution and Chemical Properties

The chemical behavior of an atom is determined by the distribution of electrons in the atom's electron shells. Beginning with hydrogen, the simplest atom, we can imagine building the atoms of the other elements by adding 1 proton and 1 electron at a time (along with an appropriate number of neutrons). **Figure 2.7**, a modified version of what is called the *periodic table of the elements*, shows this distribution of electrons for the first 18 elements, from hydrogen (H1) to argon (A18r). The elements are arranged in three rows, or *periods*, corresponding to the number of electron shells in their atoms. The left-to-right sequence of elements in each row corresponds to the sequential addition of electrons and protons. (See the complete periodic table at the back of the book.)

Figure 2.7

Electron distribution diagrams for the first 18 elements in the periodic table.

Inset of Helium: periodic table information shown with 2 (representing the atomic number), is above He (element symbol), which is above 4.003 (atomic mass).

In a standard periodic table (see the back of the book), information for each element is presented as shown for helium in the inset. In the diagrams in this table, electrons are represented as yellow dots and electron shells as concentric circles. These diagrams are a convenient way to picture the distribution of an atom's electrons among its electron shells, but these simplified models do not accurately represent the shape of the atom or the location of its electrons. The elements are arranged in rows, each representing the filling of an electron shell. As electrons are added, they occupy the lowest available shell.

The electron distribution for the first 18 elements in the periodic table is as follows:

- First Shell: Hydrogen, sub 1 H: atom with one electron in the first electron shell; Helium, sub 2 He: atom with two electrons in the first electron shell.

- Second Shell: Lithium, sub 3 Li: atom with two electrons in the first electron shell and one electron in the second shell; Beryllium, sub 4 Be: atom with two electrons in the first electron shell and two electrons in the second shell; Boron, sub 5 B: atom with two electrons in the first electron shell and three electrons in the second shell; Carbon, sub 6 C: atom with two electrons in the first electron shell and 4 electrons in the second shell; Nitrogen, sub 7 N: atom with two electrons in the first electron shell and 5 electrons in the second shell; Oxygen, sub 8 O: atom with two electrons in the first electron shell and 6 electrons in the second shell; Fluorine, sub 9 F: atom with two electrons in the first electron shell and 7 electrons in the second shell; Neon, sub 10 Ne: atom with two electrons in the first electron shell and 8 electrons in the second shell.
- Third Shell: Sodium, sub 11 Na: atom with two electrons in the first electron shell, 8 electrons in the second shell, and 1 electron in the third shell; Magnesium, sub 12 Mg: atom with two electrons in the first electron shell, 8 electrons in the second shell, and 2 electrons in the third shell; Aluminum, sub 13 Al: atom with two electrons in the first electron shell, 8 electrons in the second shell, and 3 electrons in the third shell; Silicon, sub 14 Si: atom with two electrons in the first electron shell, 8 electrons in the second shell, and 4 electrons in the third shell; Phosphorus, sub 15 P: atom with two electrons in the first electron shell, 8 electrons in the second shell, and 5 electrons in the third shell; Sulfur, sub 16 S: atom with two electrons in the first electron shell, 8 electrons in the second shell, and 6 electrons in the third shell; Chlorine, sub 17 Cl: atom with two electrons in the first electron shell, 8 electrons in the second shell, and 7 electrons in the third shell; Argon, sub 18 Ar: atom with two electrons in the first electron shell, 8 electrons in the second shell, and 8 electrons in the third shell.

VISUAl SKILLS*What is the atomic number of magnesium? How many protons and electrons does it have? How many electron shells? How many valence electrons?*

*For suggested answer, see*Appendix A.**

Animation

Electron Distribution Diagrams

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Hydrogen's 1 electron and helium's 2 electrons are located in the first shell. Electrons, like all matter, tend to exist in the lowest available state of potential energy. In an atom, this state is in the first shell. However, the first shell can hold no more than 2 electrons; thus, hydrogen and helium are the only elements in the first row of the table. In an atom with more than 2 electrons, the additional electrons must occupy higher shells because the first shell is full. The next element, lithium, has 3 electrons. Two of these electrons fill the first shell, while the third electron occupies the second shell. The second shell holds a maximum of 8 electrons. Neon, at the end of the second row, has 8 electrons in the second shell, giving it a total of 10 electrons.

The chemical behavior of an atom depends mostly on the number of electrons in its *outermost* shell. We call those outer electrons **valence electrons** and the outermost electron shell the **valence shell**. In the case of lithium, there is only 1 valence electron, and the second shell is the valence shell. Atoms with the same number of electrons in their valence shells exhibit similar chemical behavior. For example, fluorine (F) and chlorine (Cl) both have 7 valence electrons, and both form compounds when combined with the element sodium (Na): Sodium fluoride (NaF) is commonly added to toothpaste to prevent tooth decay, and, as described earlier, NaCl is table salt (see Figure 2.2). An atom with a completed valence shell is unreactive; that is, it will not interact readily with other atoms. At the far right of the periodic table are helium, neon, and argon, the only three elements shown in Figure 2.7 that have full valence shells. These elements are said to be *inert*, meaning chemically unreactive. All the other atoms in Figure 2.7 are chemically reactive because they have incomplete valence shells.

Electron Orbitals

In the early 1900s, the electron shells of an atom were visualized as concentric paths of electrons orbiting the nucleus, somewhat like planets orbiting the sun. It is still convenient to use two-dimensional concentric-circle diagrams, as in Figure 2.7, to symbolize three-dimensional electron shells. However, you need to remember that each concentric circle represents only the *average* distance between an electron in that shell and the nucleus. Accordingly, the concentric-circle diagrams do not give a real picture of an atom. In reality, we can never know the exact location of an electron. What we can do instead is describe the space in which an electron spends most of its time. The three-dimensional space where an electron is found 90% of the time is called an **orbital**.

Each electron shell contains electrons at a particular energy level, distributed among a specific number of orbitals of distinctive shapes and orientations. **Figure 2.8** shows the orbitals of neon as an example, with its electron distribution diagram for reference. You can think of an orbital as a component of an electron shell. The first electron shell has only one spherical s orbital (called 1s), but the second shell has four orbitals: one large spherical s orbital (called 2s) and three dumbbell-shaped p orbitals (called 2p orbitals). (The third shell and other higher electron shells also have s and p orbitals, as well as orbitals of more complex shapes.)

Figure 2.8

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Electron orbitals.
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Electron orbital diagrams are given.

Figure a. Electron distribution diagram. An electron distribution diagram is shown here for a neon atom, which has a total of 10 electrons. Each concentric circle represents an electron shell, which can be subdivided into electron orbitals. The first shell has 2 electrons, and the second shell has 8.

Figure b. Separate electron orbitals. The three-dimensional shapes represent electron orbitals - the volumes of space where the electrons of an atom are most likely to be found. Each orbital holds a maximum of 2 electrons. The first electron shell has one spherical s orbital, designated 1 s. The second shell has one large s orbital, designated 2 s for the second shell, plus three dumbbell-shaped orbitals called p orbitals (2 p for the second shell. The three 2 p orbitals lie at right angles to one another along imaginary x, y, and z axes of the atom. That is, dumbbell shapes extend along axes from the origin of an x y z plane.

Figure c. Superimposed electron orbitals. To reveal the complete picture of the electron orbitals of neon, we superimpose the 1 s orbital of the first shell and the 2 s and three 2 p orbitals of the second shell. Concentric circles overlap the dumbbell shapes.

No more than 2 electrons can occupy a single orbital. The first electron shell can therefore accommodate up to 2 electrons in its s orbital. The lone electron of a hydrogen atom occupies the 1s orbital, as do the 2 electrons of a helium atom. The four orbitals of the second electron shell can hold up to 8 electrons, 2 in each orbital. Electrons in each of the four orbitals in the second shell have nearly the same energy, but they move in different volumes of space.

The reactivity of an atom arises from the presence of unpaired electrons in one or more orbitals of the atom's valence shell. As you will see in the next section, atoms interact in a way that completes their valence shells. When they do so, it is the *unpaired* electrons that are involved.